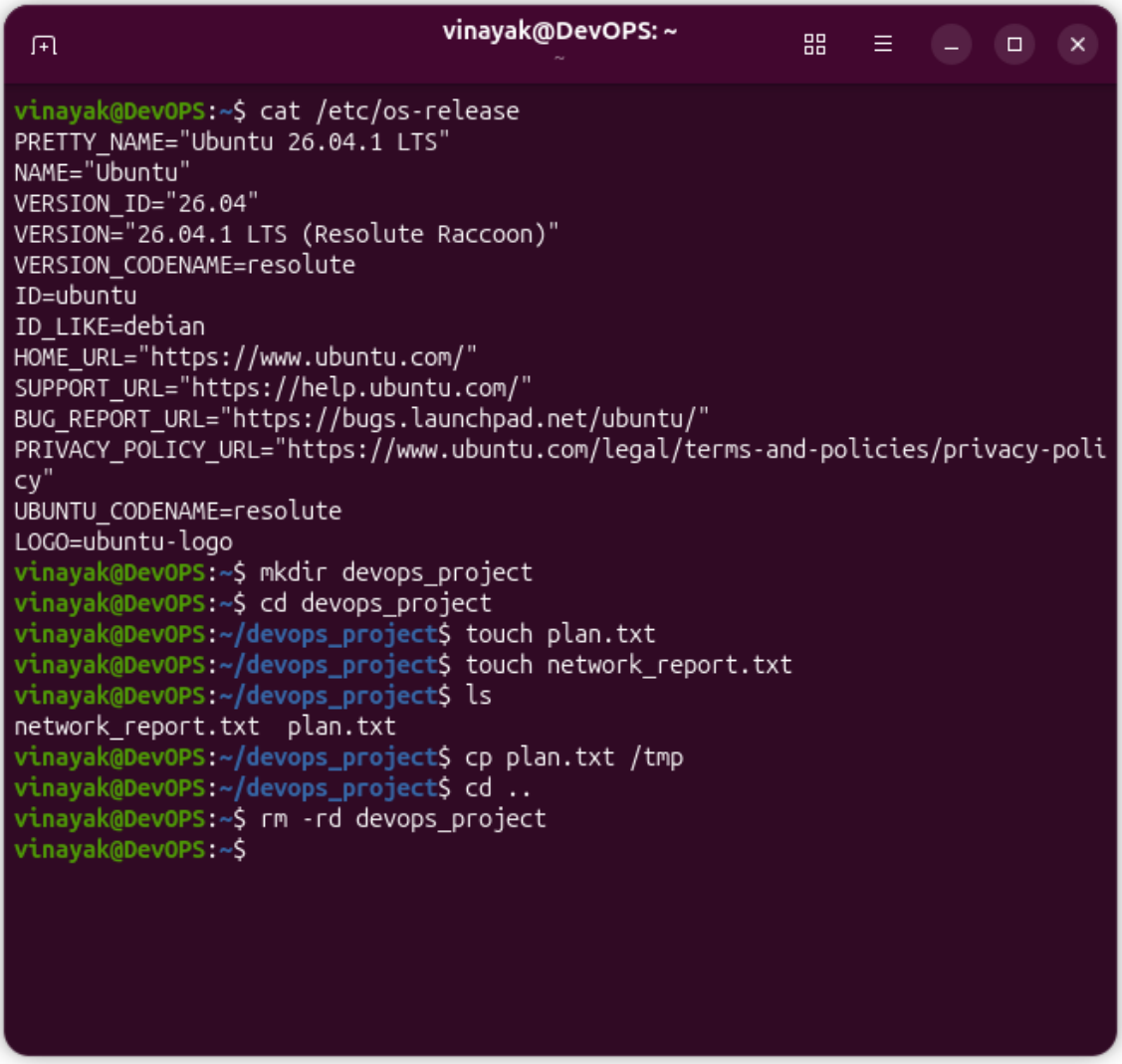
EMC Assignment 1 – Karpaga Vinayagam Vengatesan

1. Run a command to check the Linux distribution name and version of the operatin g system installed on your machine.

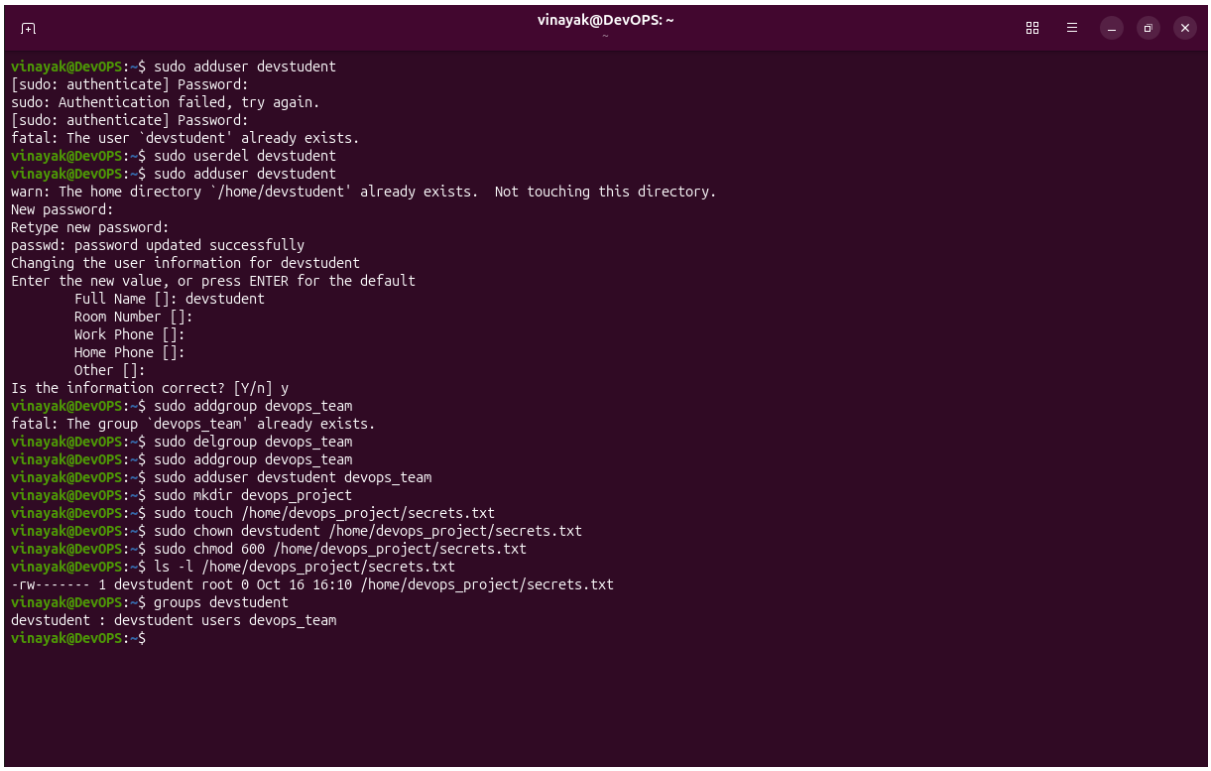
Files & Directories:

- Create directory /home/devops_project
- Inside it, create 2 files: plan.txt and network_report.txt
- Copy plan.txt to /tmp and delete the original



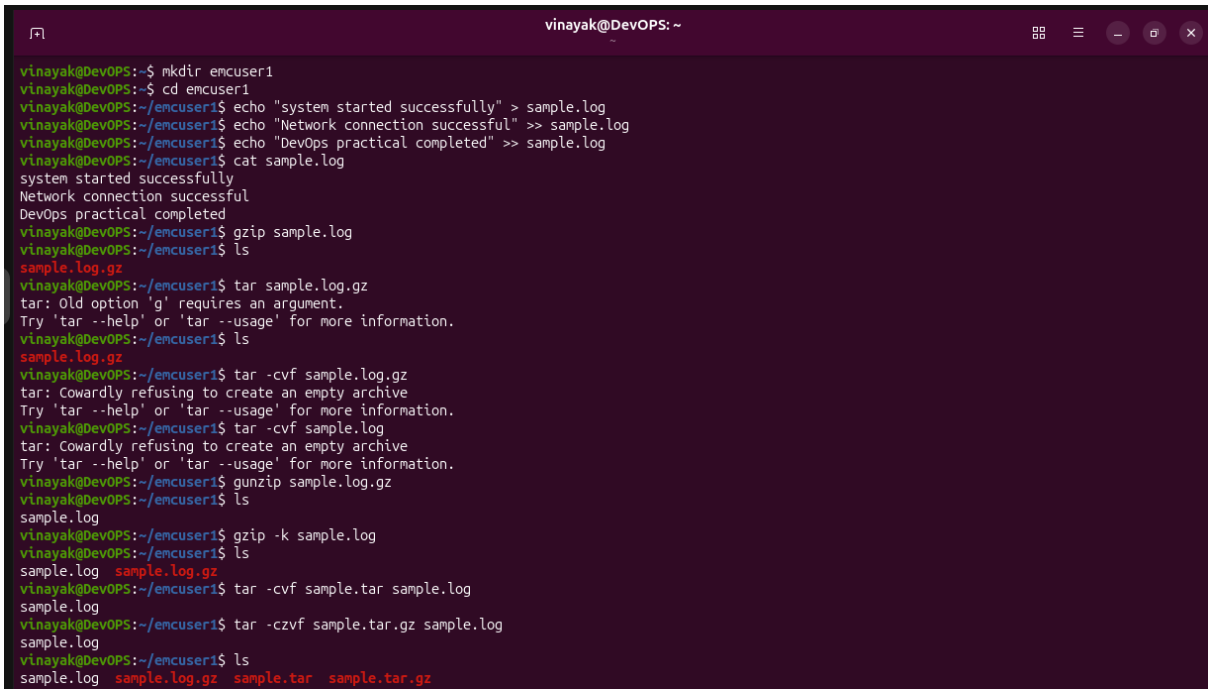
Users & Permissions:

- Create a user devstudent
- Create a group devops_team
- Add devstudent to devops_team
- Create a file secrets.txt, and:
- Change its owner to devstudent
- Set permissions so only owner can read/write



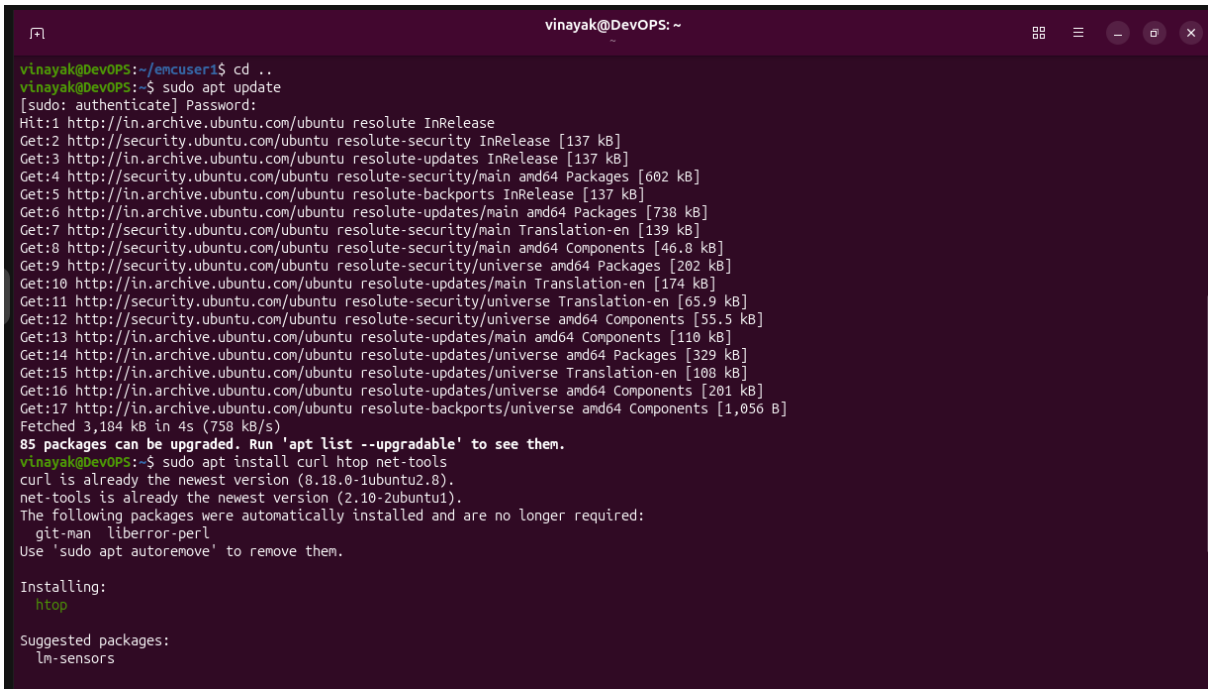
File Compression:

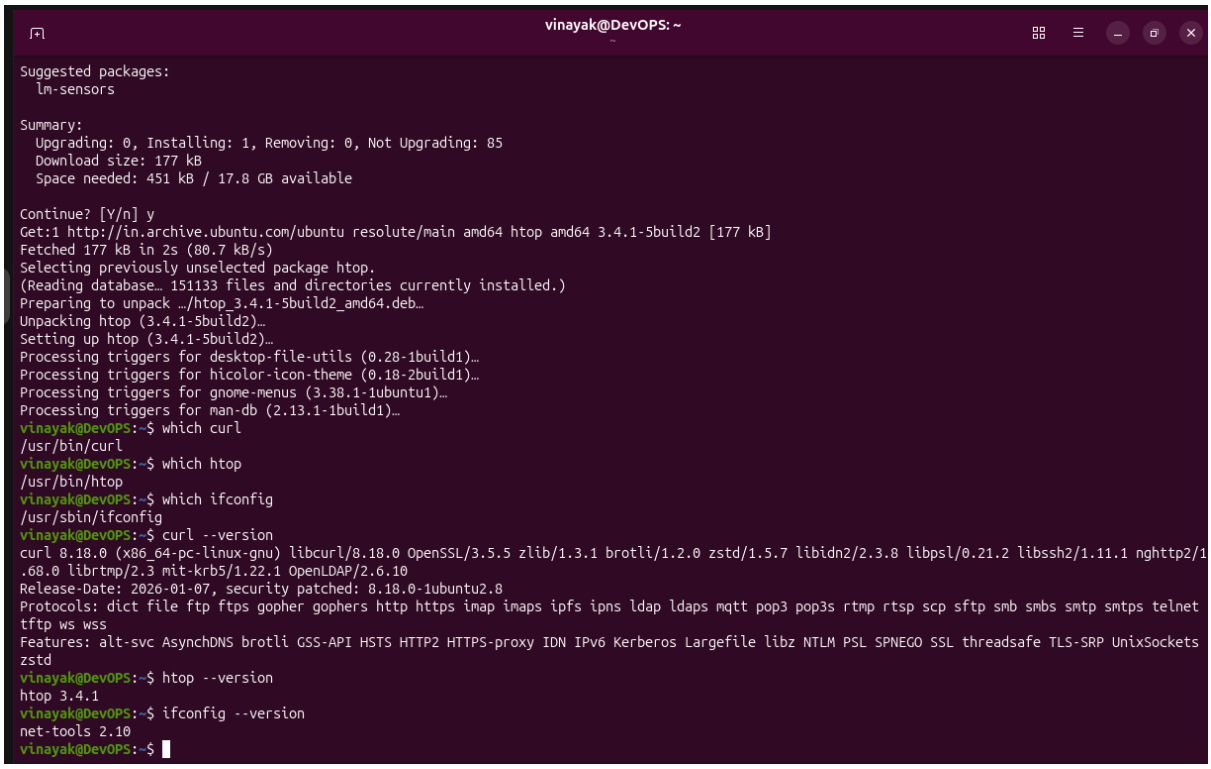
- Download a sample .log file from internet or create dummy one using echo
- Compress it using gzip and tar



Package Management:

- Install curl, htop, and net-tools using apt
- Show which and --version outputs





Run and shortly explain outputs of the following commands:

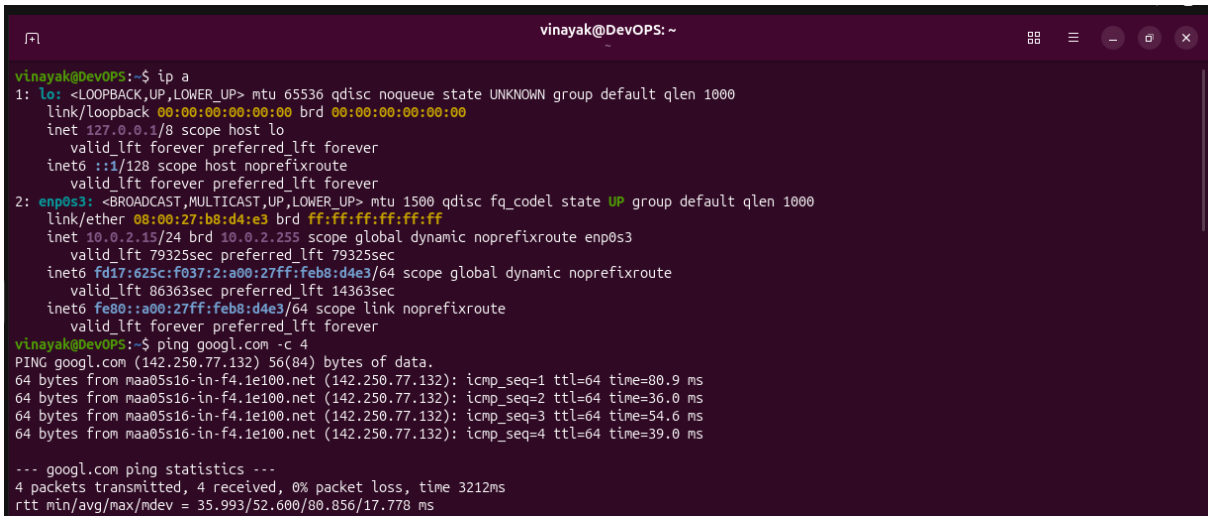
- IP a or ifconfig
- ping google.com -c 4
- traceroute google.com
- nslookup google.com
- netstat -tuln
- curl ifconfig.me

1. **IP a** – Display IP addresses

Display network interfaces, IP addresses, and their status.

2. ping google.com -c 4: Test internet connectivity

Sends four packets to Google to test network connectivity and response time.



3. Traceroute google.com – trace the network route

Shows the network hops between your ubuntu machine and google. Some hops may show if routers do not respond.

```
vinayak@DevOPS:~$ sudo apt install traceroute
The following packages were automatically installed and are no longer required:
  git-man  liberror-perl
Use 'sudo apt autoremove' to remove them.

Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 85
  Download size: 54.4 kB
  Space needed: 167 kB / 17.8 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 traceroute amd64 1:2.1.6-1build1 [54.4 kB]
Fetched 54.4 kB in 2s (33.9 kB/s)
Selecting previously unselected package traceroute.
(Reading database… 151143 files and directories currently installed.)
Preparing to unpack …/traceroute_1%3a2.1.6-1build1_amd64.deb…
Unpacking traceroute (1:2.1.6-1build1)…
Setting up traceroute (1:2.1.6-1build1)…
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (
traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6
 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto m
ode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (
traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptrac
eroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.13.1-1build1)…
vinayak@DevOPS:~$ traceroute google.com
traceroute to google.com (142.250.207.142), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  6.629 ms  6.175 ms  5.519 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 *
29  * * *
30  * * *
vinayak@DevOPS:~$
vinayak@DevOPS:~$ sudo apt install dnsutils
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
The following packages were automatically installed and are no longer required:
  git-man  liberror-perl
Use 'sudo apt autoremove' to remove them.

Upgrading:
  bind9-dnsutils  bind9-host  bind9-libs

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 82
  Download size: 1,529 kB
  Space needed: 6,144 B / 17.8 GB available

Continue? [Y/n] y
Get:1 http://security.ubuntu.com/ubuntu resolute-security/main amd64 bind9-host amd64 1:9.20.24-1ubuntu0.4 [49.7 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 bind9-dnsutils amd64 1:9.20.24-1ubuntu0.4 [163 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu0.4 [1,315 kB]
Fetched 1,529 kB in 3s (440 kB/s)
(Reading database… 151163 files and directories currently installed.)
Preparing to unpack …/bind9-host_1%3a9.20.24-1ubuntu0.4_amd64.deb…
Unpacking bind9-host (1:9.20.24-1ubuntu0.4) over (1:9.20.24-1ubuntu0.3)…
Preparing to unpack …/bind9-dnsutils_1%3a9.20.24-1ubuntu0.4_amd64.deb…
Unpacking bind9-dnsutils (1:9.20.24-1ubuntu0.4) over (1:9.20.24-1ubuntu0.3)…
Preparing to unpack …/bind9-libs_1%3a9.20.24-1ubuntu0.4_amd64.deb…
Unpacking bind9-libs:amd64 (1:9.20.24-1ubuntu0.4) over (1:9.20.24-1ubuntu0.3)…
Setting up bind9-libs:amd64 (1:9.20.24-1ubuntu0.4)…
Setting up bind9-host (1:9.20.24-1ubuntu0.4)…
Setting up bind9-dnsutils (1:9.20.24-1ubuntu0.4)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.43-2ubuntu2.4)…
vinayak@DevOPS:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
```

4. nslookup google.com – Check DNS

Finds the IP address associated with the domain name google.com using DNS.

5. netstat – tuln

Displays listening TCP and UDP ports on your machine

t – TCP

u - UDP

l – Listening ports

n – Show numeric addresses and port numbers

```
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.207.142
Name:   google.com
Address: 2404:6800:4009:800::200e

vinayak@DevOPS:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
udp        0      0 10.0.2.15:3702          0.0.0.0:*
udp        0      0 239.255.255.250:3702    0.0.0.0:*
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 127.0.0.1:323           0.0.0.0:*
udp        0      0 0.0.0.0:37774           0.0.0.0:*
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp6       0      0 fe80::a00:27ff:feb:3702 :::*
udp6       0      0 ff02::c:3702            :::*
udp6       0      0 ::1:323                 :::*
udp6       0      0 :::5353                 :::*
udp6       0      0 :::44373                :::*
vinayak@DevOPS:~$ sudo apt install net-tools
net-tools is already the newest version (2.10-2ubuntu1).
The following packages were automatically installed and are no longer required:
  git-man  liberror-perl
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 82
vinayak@DevOPS:~$ curl ifconfig.me
vinayak@DevOPS:~$ curl ifconfig.me
2409:40f4:2080:5141:51ef:36f2:aa6f:c40vinayak@DevOPS:~$
```

6. curl ifconfig.me – Find your public IP address

Displays the public IP address that the external website sees for your connection.